L01: Algorithms & Analysis

Asymptotic bounds & sums

$O(g): f \le Cg$; $\Omega(g): f \ge cg$; $\Theta(g): cg \le f \le Cg$ $(0 < c \le C, n \ge n_0)$. $\Theta(g) = O(g) \cap \Omega(g)$; bounds $\not\equiv$ best/worst input cases.

$$1 \prec \log n \prec n \prec n \log n \prec n^2 \prec 2^n \prec n!,$$
$$\textstyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2} = \Theta(n^2),$$
$$\textstyle\sum_{i=0}^{h} r^i = \frac{r^{h+1}-1}{r-1}\ (r \ne 1),$$
$$H_n = \textstyle\sum_{i=1}^{n} \frac{1}{i} = \Theta(\log n),$$
$$\textstyle\sum_{i=1}^{n} \log i = \log(n!) = \Theta(n \log n).$$

Loop templates: $\sum_{t=0}^{\lfloor \log_b n \rfloor} b^t = \Theta(n)$ $(b > 1)$; $\sum_{t=0}^{\lfloor \log_b n \rfloor} \lfloor n/b^t \rfloor = \Theta(n)$; $\sum_{i=1}^{n} (\lfloor \log_b i \rfloor + 1) = \Theta(n \log n)$. Θ **proof:** upper by max term; lower from last $\ge n/2$ terms (each $\Omega(\log n)$ for log sums).

Correctness & searching

Loop invariant: initialization $\to$ maintenance $\to$ termination; also prove progress/termination. **Linear:** best $\Theta(1)$; worst $\Theta(n)$; successful uniform average comparisons $(n+1)/2$; space $\Theta(1)$. **Binary:** sorted A; best $\Theta(1)$, worst/standard average $\Theta(\log n)$; iterative aux. space $\Theta(1)$, recursive (index-only) stack $\Theta(\log n)$.

$$m = l + \left\lfloor \tfrac{r-l}{2} \right\rfloor,$$
$$\begin{cases} l > r & \Rightarrow \text{return } -1, \\ A[m] = x & \Rightarrow \text{return } m, \\ A[m] < x & \Rightarrow l \leftarrow m+1, \\ A[m] > x & \Rightarrow r \leftarrow m-1. \end{cases}$$

Invariant: if x exists, it remains in $A[l \dots r]$; $l = r$ is one candidate, $l > r$ is empty (do not index).

Sorting

Algorithm	Best	Worst	Aux.	Stable
Bubble V4	n	n^2	1	Y
Insertion	n	n^2	1	Y
Selection	n^2	n^2	1	N
Merge	$n \log n$	$n \log n$	n	Y*

*Entries are $\Theta(\cdot)$; *merge ties: take left first.* **Invariants:** Bubble: suffix contains i largest (sorted); Insertion: processed prefix sorted; Selection: prefix contains i smallest (sorted); Merge: output prefix sorted after each extraction.

$$I = \textstyle\sum_{i<j} [A_i > A_j], \qquad T_{\text{ins}} = \Theta(n + I),$$
$$C_{\text{sel}} = \tfrac{n(n-1)}{2},$$
$$T_{\text{merge}} = 2T(n/2) + \Theta(n) = \Theta(n \log n).$$

Special: ordered segments of length k: bubble $\Theta(nk)$ worst case, at most $k-1$ whole-array passes (known condition); $[2, 3, \dots, n, 1]$: bubble $\Theta(n^2)$, insertion $\Theta(n)$. Two distinct keys can still give $I = \Theta(n^2)$. Selection's nonadjacent swap may break stability; Bubble V4 stops on a no-swap pass. Standard merge peak aux. space $\Theta(n)$ (reused buffer); L01 slides list $O(n \log n)$ for a different allocation/accounting convention.

Recursion tree & Master Method

For $T(n) = aT(n/b) + f(n)$ $(a \ge 1, b > 1)$, let $p = \log_b a$, $h = \log_b n$ (assume $n = b^h$, $T(1) = \Theta(1)$):

$$W_i = a^i f(n/b^i), \quad L = a^h = n^p,$$
$$T(n) = \textstyle\sum_{i=0}^{h-1} W_i + \Theta(n^p).$$

For $f(n) = \Theta(n^c)$: $W_i = \Theta\left(n^c (a/b^c)^i\right)$; $c < p$: leaves; $c = p$: equal levels; $c > p$: root. **Basic Master** (eventually nonnegative f, $\epsilon > 0$):

$$f = O(n^{p-\epsilon}) \Rightarrow T = \Theta(n^p),$$
$$f = \Theta(n^p) \Rightarrow T = \Theta(n^p \log n),$$
$$f = \Omega(n^{p+\epsilon}) \Rightarrow T = \Theta(f(n)).$$

Case 3 also requires $af(n/b) \le \rho f(n)$, $\rho < 1$.

Extended Master: $f = \Theta(n^c \log^k n)$ (eventually positive):

$$T(n) = \begin{cases} \Theta(n^p), & c < p, \\ \Theta(n^c \log^k n), & c > p, \\ \Theta(n^p \log^{k+1} n), & c = p,\ k > -1, \\ \Theta(n^p \log\log n), & c = p,\ k = -1, \\ \Theta(n^p), & c = p,\ k < -1. \end{cases}$$

At $c = p$, $h = \log_b n$:

$$T = \Theta\left(n^p \textstyle\sum_{j=1}^{h} j^k\right).$$

$\sum j^{-1} = \Theta(\log h)$, $\sum j^{-2} = \Theta(1)$.

L01 exercise traps

Signed recurrence: standard Master does not apply to $f(n) = n \sin n$. For $n = 2^h$, $T(1) = c > 0$:

$$T(n) = n^2 \left(c + \textstyle\sum_{j=1}^{h} \frac{\sin(2^j)}{2^j}\right) = \Theta(n^2).$$

Indeed $S_2 = \sin 2/2 + \sin 4/4 \approx 0.26545 > 1/4 = \sum_{j \ge 3} 2^{-j}$, hence $0 \le S_h \le 1$. Negative terms can cancel: $4T(n/2) - n$, $T(1) = 1$ has solution $T(n) = n$.

L02: Memory Hierarchy & Cache

Memory, locality & units

M: capacity; $\mathcal{B}$: line size; W: work; Q: misses; #lines $= M/\mathcal{B}$. Bytes for mapping, elements for matrices (e bytes/element): $(M, \mathcal{B}) = (M_{\text{bytes}}/e, \mathcal{B}_{\text{bytes}}/e)$. **Latency:** time/op; data throughput $\le$ bandwidth; high bandwidth $\not\Rightarrow$ low latency. **Locality:** temporal = same soon; spatial = nearby soon. **CPU:** registers $\to$ L1 $\to$ L2 $\to$ L3 cache $\to$ DRAM; L1/L2 are on-chip: L1 closest to each core; L2 per core or core cluster. L3 typically shared across cores. Hardware manages caches.

GPU (CUDA): SM = streaming multiprocessor. Inside each SM: registers (thread-private), shared memory (per thread block; managed by code), L1 cache (managed by hardware). L2: on-chip, outside individual SMs, shared across all SMs. Global memory: device DRAM accessible to all threads; local memory: thread-private address space backed by device DRAM.

FPGA: on-chip: registers, LUTRAM, BRAM/URAM; off-chip: DRAM. Designer chooses buffer layout and dataflow.

Cache mapping

Byte address a, w bits, k ways; M excludes metadata; $\mathcal{B}$, E powers of two.

$$E = \frac{M}{k\mathcal{B}} \text{ sets}, \quad q = \lfloor a/\mathcal{B} \rfloor,$$
$$o = a \bmod \mathcal{B}, \quad s = q \bmod E,$$
$$t = \lfloor q/E \rfloor = \lfloor a/(M/k) \rfloor,$$
$$a = [\text{tag} \mid \text{set index} \mid \text{offset}],$$
$$b_o = \log_2 \mathcal{B}, \quad b_s = \log_2 E,$$
$$b_t = w - b_s - b_o = w - \log_2(M/k).$$

DM (direct): $k = 1$; **k-way:** any way in set s; **FA (fully assoc.):** $k = M/\mathcal{B}$, $E = 1$, $b_s = 0$.

$$\text{Hit} \iff \exists j \text{ in set } s:\ v_j = 1 \wedge t_j = t.$$

LRU: hit/fill $\to$ MRU; miss $\to$ invalid way first, else evict LRU within set.

Miss types

Actual miss; reference = same-capacity FA-LRU, same trace/$\mathcal{B}$; ignore coherence:

Cold	first access to block
Capacity	seen; reference misses
Conflict	seen; reference hits

Conflict: DM and k-way; FA has no mapping conflicts. Full set $\not\Rightarrow$ capacity miss. **Coherence (writes):** true sharing = same data; false sharing = different data, same line.

Blocking / tiling

Supplementary derivation. $S \times S$ tiles; FA; $\mathcal{B} \le S \le n$, $S \mid n$; three tile line footprints fit. Keep C_{IJ} resident across inner K: $C_{IJ} \mathrel{+}= A_{IK} B_{KJ}$.

$$W_{\text{tile}} = \Theta(S^3), \quad D_{\text{tile}} = \Theta(S^2),$$
$$Q_{\text{tile}} = O(S^2/\mathcal{B} + S) = O(S^2/\mathcal{B}),$$
$$\#\text{tile products} = (n/S)^3,$$
$$Q_{A+B} = O(n^3/(\mathcal{B}S)), \quad Q_C = O(n^2/\mathcal{B}),$$
$$Q_{\text{blocked}} = O\left(\frac{n^3}{\mathcal{B}S} + \frac{n^2}{\mathcal{B}}\right).$$

$3S^2 \lesssim M$; large n: choose $S = \Theta(\sqrt{M})$ with line-boundary headroom (tall cache).

$$Q_{\text{blocked}} = O\left(\frac{n^3}{\mathcal{B}\sqrt{M}} + \frac{n^2}{\mathcal{B}}\right).$$

$W(n) = \Theta(n^3)$ unchanged; work/data $= \Theta(S)$.

Double buffering

$N \ge 1$ equal chunks; $T_L, T_C > 0$: load/compute per chunk. Two buffers; load $\|$ compute; ignore sync/switch/output costs.

$$H = \max(T_L, T_C), \quad L = \min(T_L, T_C),$$
$$T_{\text{single}} = N(T_L + T_C),$$
$$T_{\text{double}} = T_L + (N-1)H + T_C = NH + L,$$
$$\text{SU}_N = \frac{N(H+L)}{NH+L},$$
$$\text{SU}_\infty = 1 + \frac{L}{H} \le 2,$$
$$T_L = T_C \Rightarrow \text{SU}_N = \frac{2N}{N+1} < 2.$$

$N = 1$: SU $= 1$; $T_L > T_C$: load-bound; $T_C > T_L$: compute-bound.

Ideal cache & lemmas

FA; **Ideal** = optimal replacement. LRU comparison: same trace/$\mathcal{B}$, cold caches.

$$Q_{\text{LRU}}(2M) \le 2Q_{\text{Ideal}}(M).$$

Cache-Miss: access only r contiguous segments; $N = \sum_i S_i$; FA LRU/ideal.

$$N < M/3, \quad N/r \ge \mathcal{B} \Longrightarrow Q \le N/\mathcal{B} + 2r \le 3N/\mathcal{B}.$$

Tall cache: $\mathcal{B}^2 < cM$ $(c > 0$ suitably small); $M = \Omega(\mathcal{B}^2)$. **Submatrix:** row-major $n \times n$ $(N = n^2, r = n)$:

$$\mathcal{B}^2 < cM \le n^2 < M/3 \Longrightarrow Q \le 3n^2/\mathcal{B}.$$

Sufficient conditions; upper bounds $\not\equiv$ exact counts.

Naive matrix multiplication

Row-major, $i \to j \to k$; FA-LRU, tall cache, cold start; M, $\mathcal{B}$ in elements.

$$C_{ij} \mathrel{+}= A_{ik} B_{kj}, \qquad W(n) = \Theta(n^3).$$

k-loop: A contiguous; B stride n; C_{ij} reused.

Size condition	$Q(n)$
$n > c_1 M/\mathcal{B}$	$\Theta(n^3)$
$c_2\sqrt{M} < n < c_3 M/\mathcal{B}$	$\Theta(n^3/\mathcal{B})$
$n < c_4\sqrt{M}$	$\Theta(1 + n^2/\mathcal{B})$

c_i: suitable constants; transition bands omitted. **Fit scales:** $n\mathcal{B} \lesssim M$: one B line/row; $3n^2 \lesssim M$: all A, B, C (line overhead omitted). Middle: reuse across $\mathcal{B}$ adjacent columns; small: across all i.